FitHub CRM

Fitness Center Management

Functional & Performance Testing

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9. Functional & Performance Testing

9.1 Functional Testing Overview

Functional testing ensures that every feature of the FitHub CRM application works as specified in the requirements. Testing was conducted in a Salesforce Developer Org using manual test cases covering all functional requirements. The goal of this phase is to validate that the application behaves correctly under expected conditions and appropriately handles edge cases.

Testing Approach:

- Manual Testing in Salesforce Developer Org
- Test cases mapped to Functional Requirements (FR-1 through FR-8)
- All critical business processes validated
- Edge cases and error scenarios tested

9.2 Test Cases

TC ID	Module/Feature	Test Scenario	Expected Result	Actual Result	Status
TC-01	Member	Create new member with all required fields	Record created with auto-number ID (MEM-XXXXXX)	Success - Auto-number ID generated	✅ Pass

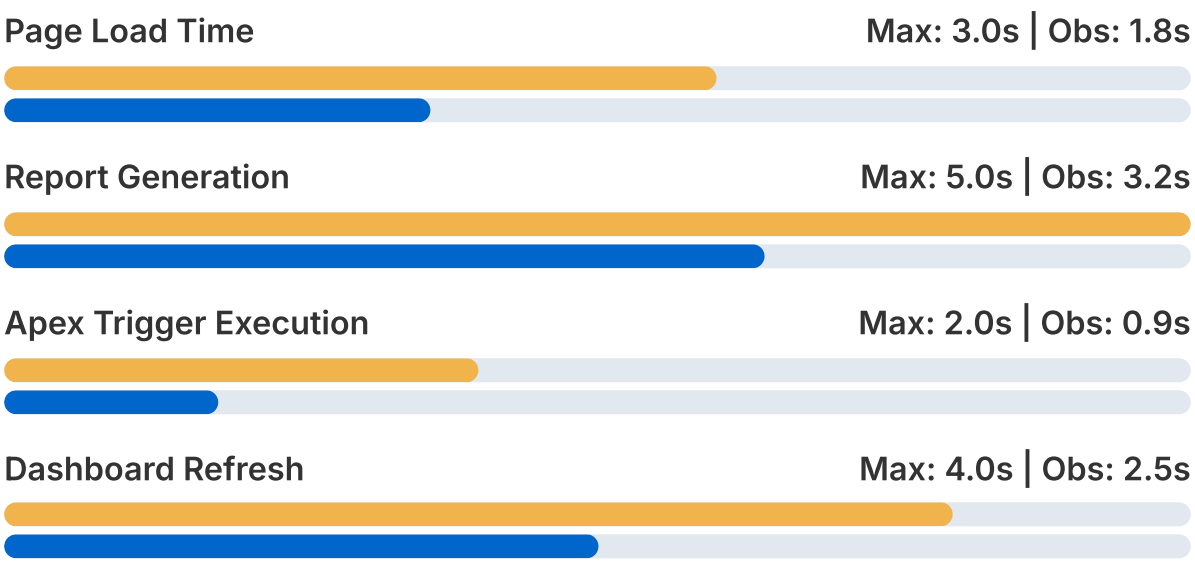
TC ID	Module/Feature	Test Scenario	Expected Result	Actual Result	Status
TC-02	Member	Set subscription end date before start date	Validation error should be displayed	Validation error displayed correctly	✅ Pass
TC-03	Class/Booking	Add booking when class is at full capacity	Overbooking should be blocked by validation rule	Overbooking blocked - Error: "Class is already full"	✅ Pass
TC-04	Booking	Confirm booking without marking payment as Paid	Confirmation should be blocked; error message shown	Confirmation blocked - Payment validation enforced	✅ Pass
TC-05	Trainer/Class	Create a class with matching trainer specialization	Trainer should be auto-assigned via Apex trigger	Trainer auto-assigned successfully	✅ Pass
TC-06	Flow	Update membership status to Expired	Expiry email should be sent automatically	Expiry email sent via Flow	✅ Pass
TC-07	Scheduled Apex	Subscription ending in 5 days	Pre-expiry reminder email should be sent	Reminder email sent by scheduled job	✅ Pass
TC-08	Reports	View FitHub Analysis Report	All records with class, booking, and amount data visible	Report generated with accurate data	✅ Pass

TC ID	Module/Feature	Test Scenario	Expected Result	Actual Result	Status
TC-09	Dashboard	Open FitHub Dashboard	Charts should render real-time data	Dashboard charts rendered correctly	✅ **Pass**
TC-10	Security	Login as Front Desk user and access Trainer records	Read-only access should be enforced	Read-only access enforced correctly	✅ **Pass**

9.3 Performance Testing

Test Area	Benchmark	Observed Result	Status
Page Load Time	< 3 seconds	~1.8 seconds	✅ Pass
Report Generation	< 5 seconds	~3.2 seconds	✅ Pass
Apex Trigger Execution	< 2 seconds per DML	~0.9 seconds	✅ Pass
Scheduled Apex Job	Runs at 6 AM daily	Executed on schedule	✅ Pass
Email Delivery (Flow)	< 1 minute	~30 seconds	✅ Pass
Dashboard Refresh	< 4 seconds	~2.5 seconds	✅ Pass

Visual Performance Chart (Benchmark vs. Observed)



Benchmark Limit Observed Time

9.4 Test Summary

Total Test Cases:	10	**Passed:**	10
Failed:	0	**Pass Rate:**	100%

✅ **ALL TESTS PASSED**

Testing Conclusion:
All functional and performance test cases have passed successfully. The FitHub CRM application meets all specified requirements and performs well within acceptable benchmarks. System stability and feature integrity have been thoroughly verified. The system is fully operational and is now ready for User Acceptance Testing (UAT) and production deployment.